1.4 - The Matrix Equation $A\mathbf{x} = \mathbf{b}$

Notes: Key exercises are 1–20, 37, 38, 41 and 42. Exercises 39, 40, 43, and 44 are harder. Exercise 44 anticipates the Invertible Matrix Theorem but is not used in the proof of that theorem.

9. The system has the same solution set as the vector equation

$$x_1\begin{bmatrix}3\\0\end{bmatrix}+x_2\begin{bmatrix}1\\1\end{bmatrix}+x_3\begin{bmatrix}-5\\4\end{bmatrix}=\begin{bmatrix}9\\0\end{bmatrix}$$

and this equation has the same solution set as the matrix equation

$$\begin{bmatrix}3&1&-5\\0&1&4\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}9\\0\end{bmatrix}$$

11. To solve $A\mathbf{x} = \mathbf{b}$, row reduce the augmented matrix $[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{b}]$ for the corresponding linear system:

$$\begin{bmatrix}1&2&4&-2\\0&1&5&2\\-2&-4&-3&9\end{bmatrix}\sim\begin{bmatrix}1&2&4&-2\\0&1&5&2\\0&0&5&5\end{bmatrix}\sim\begin{bmatrix}1&2&4&-2\\0&1&5&2\\0&0&1&1\end{bmatrix}\sim\begin{bmatrix}1&2&0&-6\\0&1&0&-3\\0&0&1&1\end{bmatrix}\sim\begin{bmatrix}\textcircled{1}&0&0&0\\0&\textcircled{1}&0&-3\\0&0&\textcircled{1}&1\end{bmatrix}$$

The solution is $\begin{cases}x_1 = 0\\x_2 = -3\\x_3 = 1\end{cases}$. As a vector, the solution is $\mathbf{x} = \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}0\\-3\\1\end{bmatrix}$.

13. The vector $\mathbf{u}$ is in the plane spanned by the columns of A if and only if $\mathbf{u}$ is a linear combination of the columns of A. This happens if and only if the equation $A\mathbf{x} = \mathbf{u}$ has a solution. (See the box preceding Example 3 in Section 1.4.) To study this equation, reduce the augmented matrix $[A \quad \mathbf{u}]$

$$\begin{bmatrix}3&-5&0\\-2&6&4\\1&1&4\end{bmatrix}\sim\begin{bmatrix}1&1&4\\-2&6&4\\3&-5&0\end{bmatrix}\sim\begin{bmatrix}1&1&4\\0&8&12\\0&-8&-12\end{bmatrix}\sim\begin{bmatrix}\textcircled{1}&1&4\\0&\textcircled{8}&12\\0&0&0\end{bmatrix}$$

The equation $A\mathbf{x} = \mathbf{u}$ has a solution, so $\mathbf{u}$ is in the plane spanned by the columns of A.

For your information: The unique solution of $A\mathbf{x} = \mathbf{u}$ is (5/2, 3/2).

15. The augmented matrix for $A\mathbf{x} = \mathbf{b}$ is $\begin{bmatrix}2&-1&b_1\\-6&3&b_2\end{bmatrix}$, which is row equivalent to $\begin{bmatrix}\textcircled{2}&-1&b_1\\0&0&b_2+3b_1\end{bmatrix}$. This shows that the equation $A\mathbf{x} = \mathbf{b}$ is not consistent when $3b_1 + b_2$ is nonzero. The set of $\mathbf{b}$ for which the equation *is* consistent is a line through the origin – the set of all points (b_1, b_2) satisfying $b_2 = -3b_1$.